

LeoEM ISL Routing Emulation and Analysis

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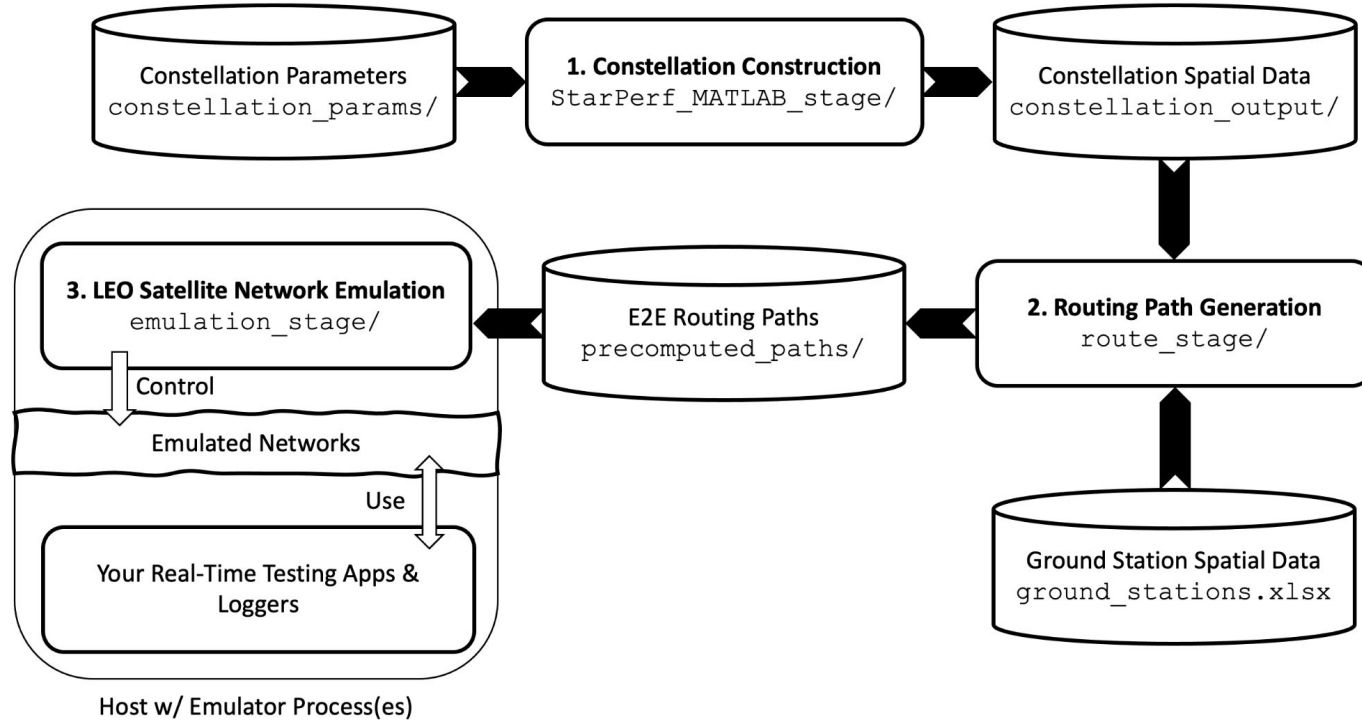
Overview

- Background
- Methodology
- TCP Result
- SaTCP Result
- Comparison
- Conclusion and Future Improvements

Background

- Challenges in LEO Communication
 - Frequent Handover Events
 - High Round-Trip Time (RTT) Variability
 - Throughput Fluctuations
- SaTCP
 - TCP adaptation designed for LEO networks

Emulation Stages [1]



[1] <https://github.com/XuyangCaoUCSD/LeoEM/blob/main/workflow.png>

Stage 1

- Constellation Spatial Data
 - Using Starlink data
 - Satellite number: 1588
 - Cycles: 5731

Stage 2

- Route computation
 - Using inter-satellite link (ISL)
 - precomputed route from SD to Shanhei

Stage 3

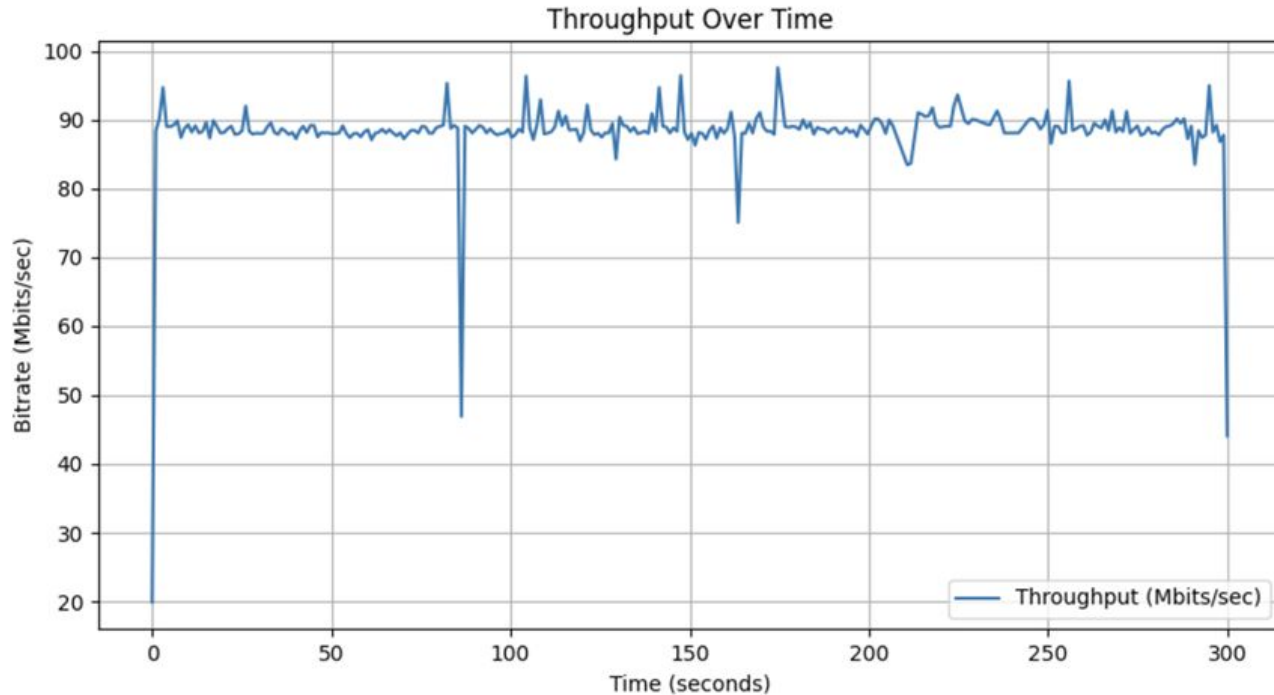
- Emulation
 - Testing throughput and RTT from two endpoints
 - Testing with TCP
 - Setting up and testing with SaTCP

Emulation

- Setup
 - Ubuntu Server 24.04 on VM
 - Using Starlink_SD_Shanghai_15_ISL_path.log as route
 - Test throughput using iPerf3
 - h1(10.0.1.101) as client
 - h2(10.0.1.102) as server
 - Test RTT using tshark to capture iPerf traffic
 - Log handover time for analysis

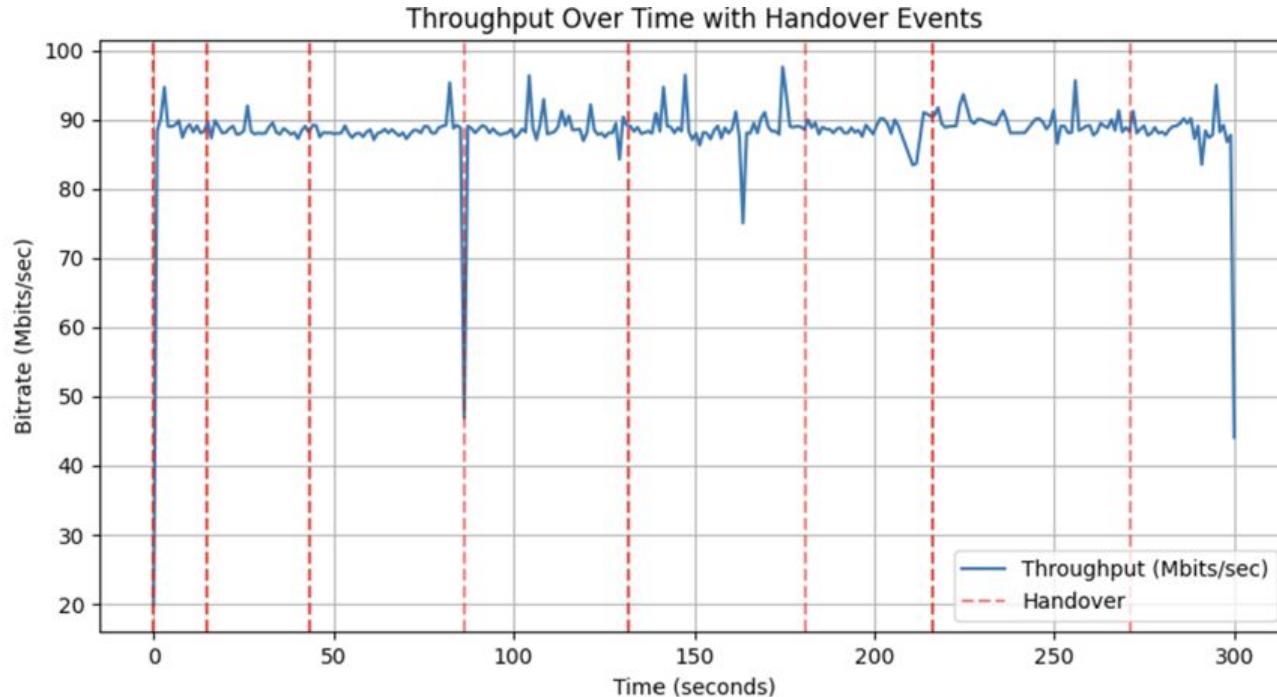
Emulation Result (TCP)

- Throughput result



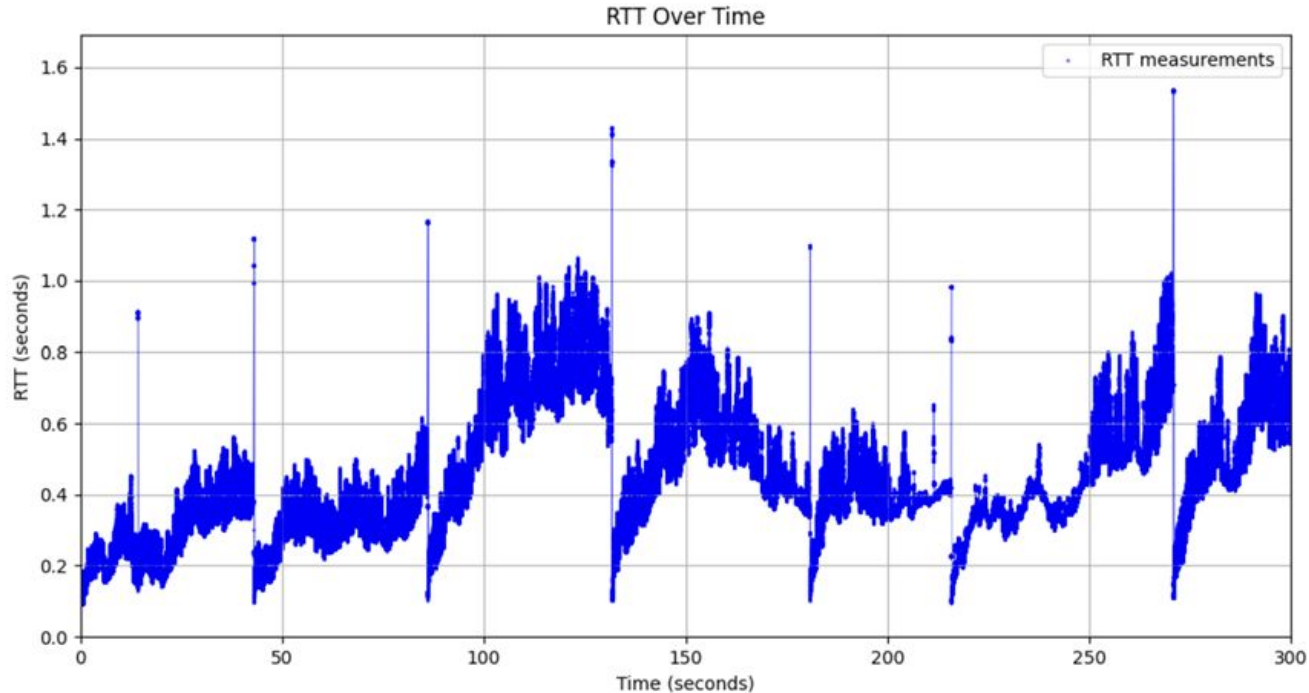
Emulation Result (TCP)

- Throughput result with handover events



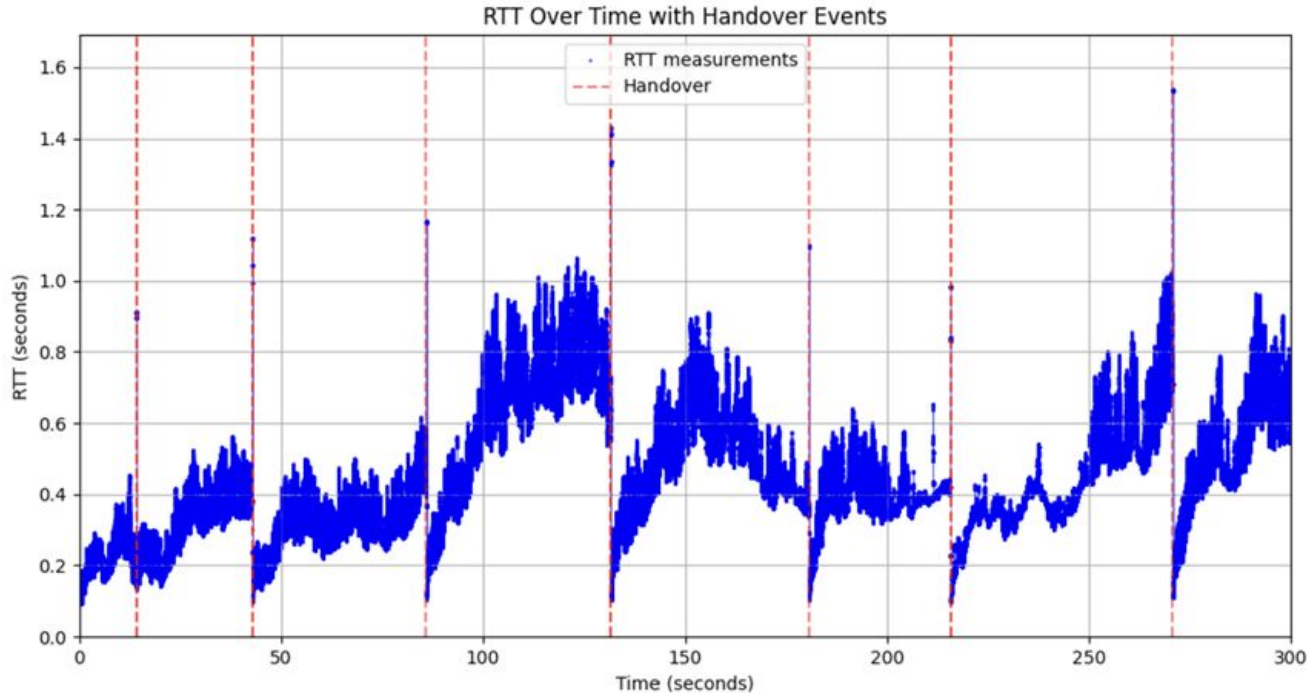
Emulation Result (TCP)

- RTT result



Emulation Result (TCP)

- RTT result with handover events

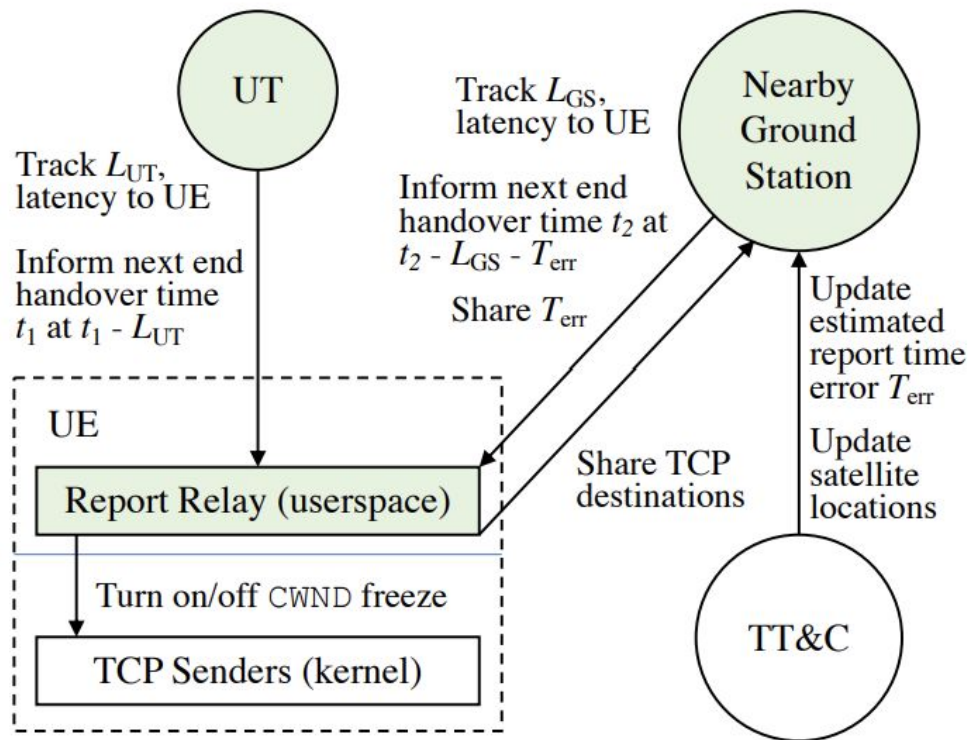


SaTCP

- Features of SaTCP [2]
 - Prediction of satellite location
 - Compute when a disruptive event will occur
 - Freeze congestion window during handover
 - Cross-Layer Integration
 - Collect real-time data from the link layer

SaTCP

- Workflow [3]

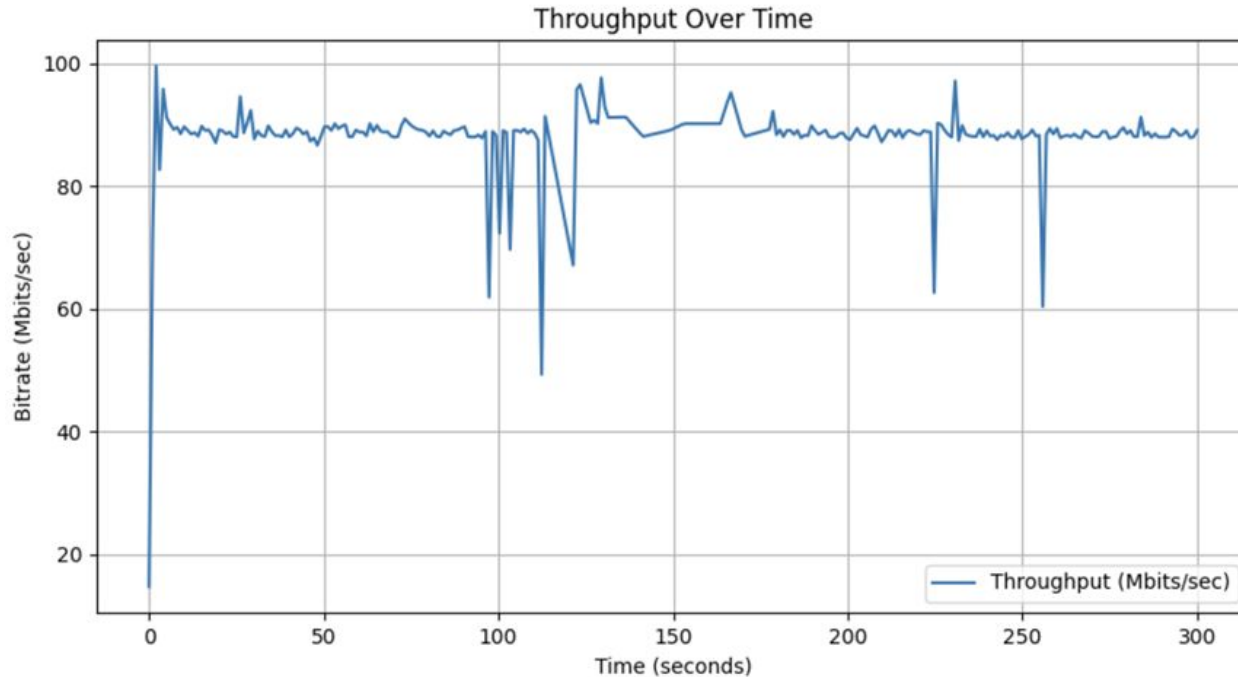


TCP to SaTCP

- Setup
 - Install Linux kernel source code 6.9.6 on VM
 - Follow the instructions of SaTCP Implementation in LeoEM
 - Run the emulation and tests

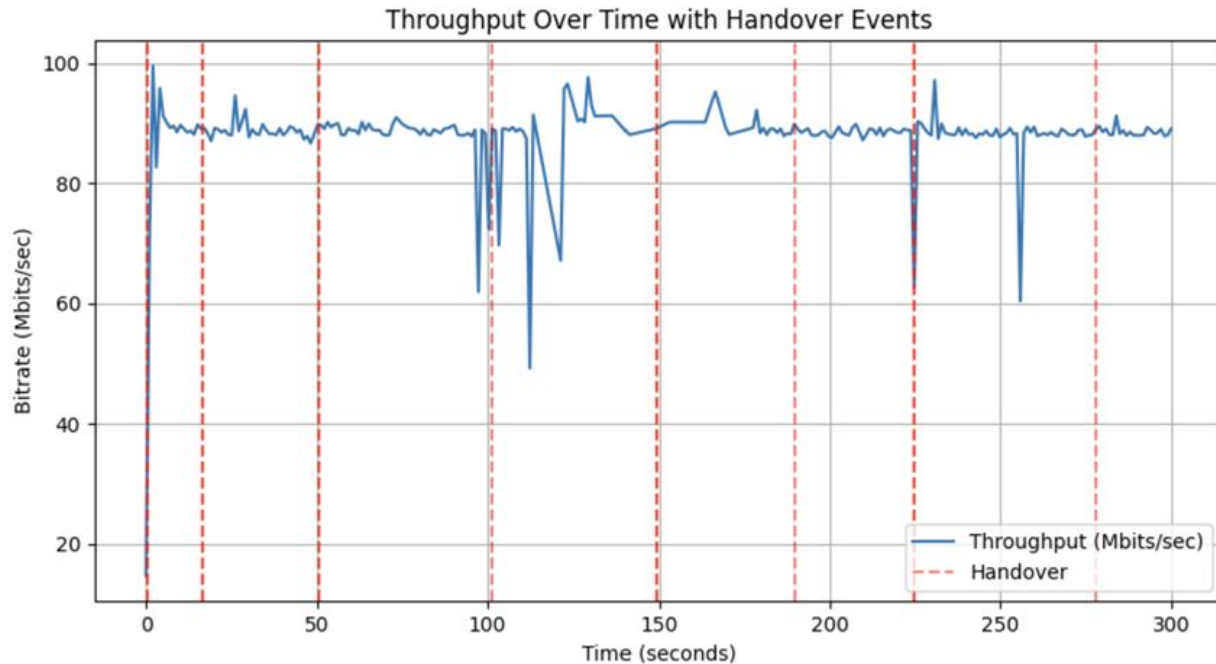
Emulation Result (SaTCP)

- Throughput result



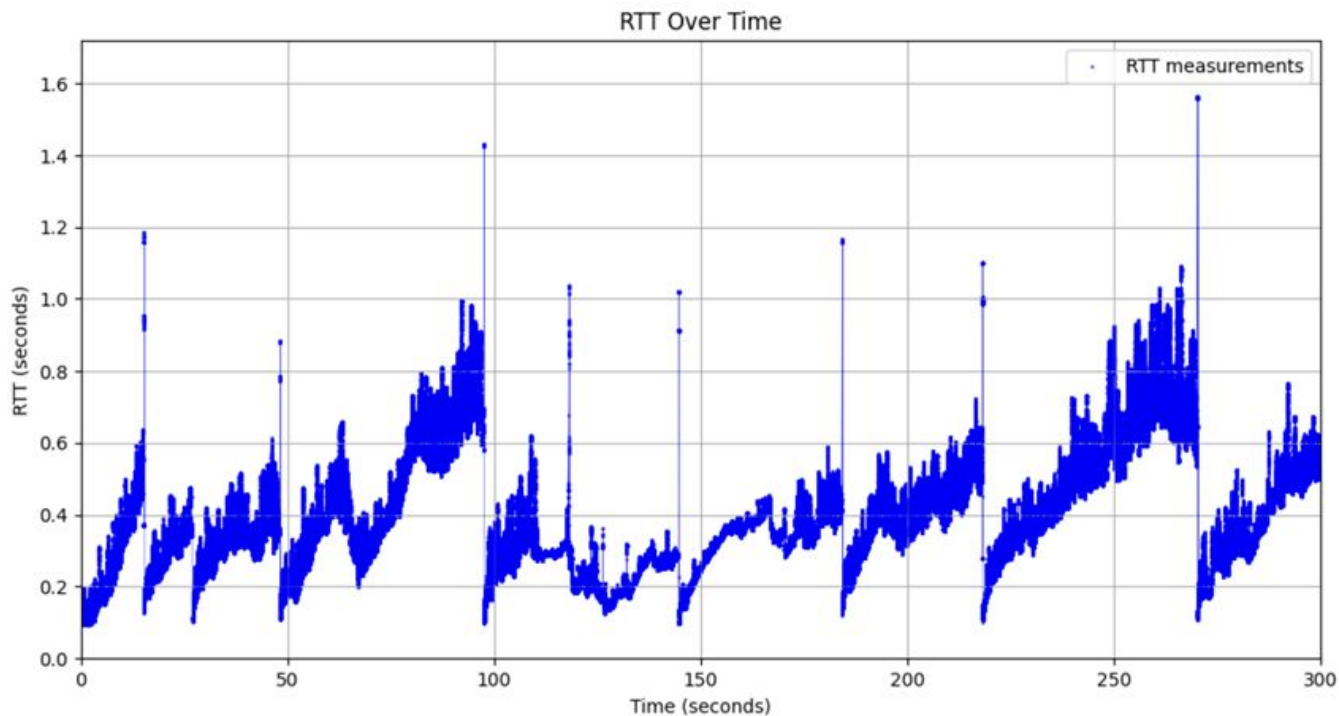
Emulation Result (SaTCP)

- Throughput result with handover events



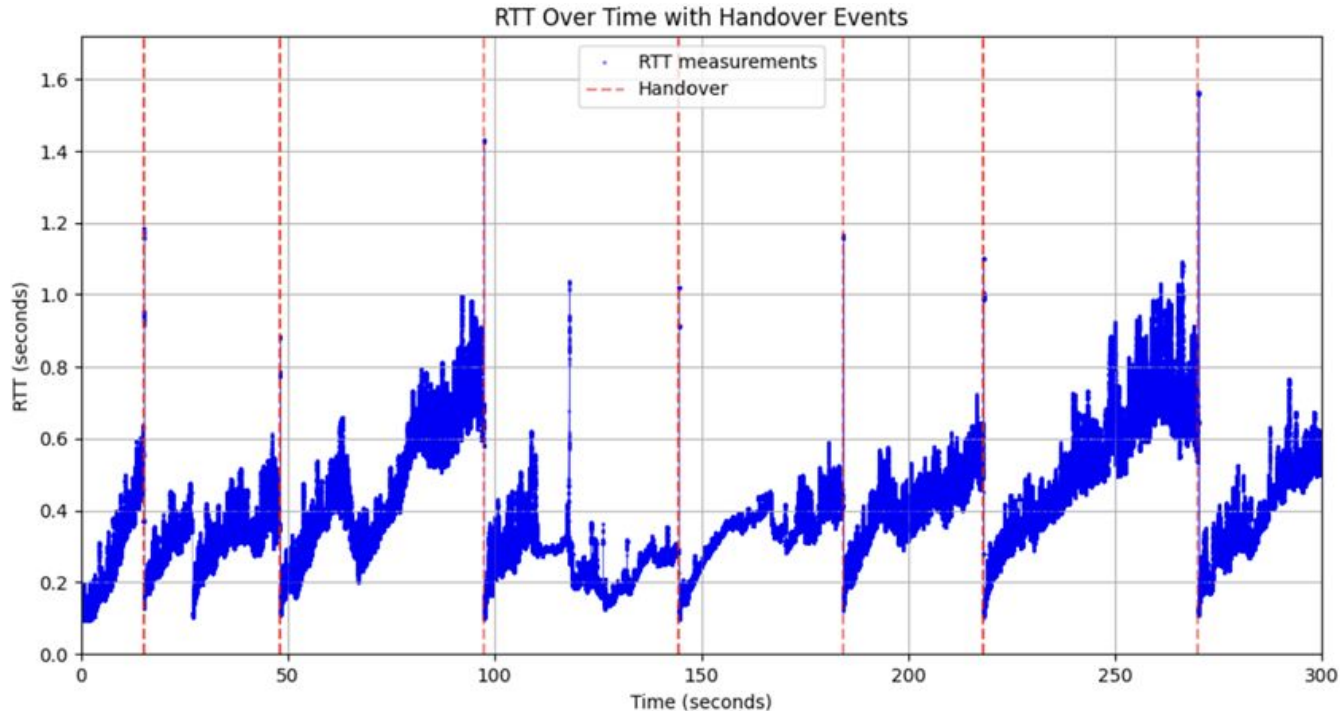
Emulation Result (SaTCP)

- RTT result



Emulation Result (SaTCP)

- RTT result with handover events



Comparison

	TCP	SaTCP
Average Throughput (Mbits/sec)	89.84	88.05
MinimumThroughput (Mbits/sec)	47.94	14.70
Average RTT (sec)	0.44	0.38
Maximum RTT (sec)	1.54	1.56

Comparison

- Throughput Comparison
 - Both had similar average throughput
 - Minimum throughput for SaTCP dropped significantly
- RTT Comparison
 - SaTCP achieved better RTT stability
 - RTT fluctuations during handovers were reduced compared to TCP

Conclusion

- Performance of SaTCP
 - Met expectations for RTT stability
 - Did not maintain consistent throughput as suggested
- Possible reasons
 - Results were influenced by resource limitations and inconsistent timing in the VM
 - SaTCP's predictive mechanisms and cross-layer data collection may introduce overhead that affects throughput

Future Improvements

- Enhance Emulation Environment
 - Use dedicated physical machines instead of VMs to minimize resource contention and timing inconsistencies
- Develop Improved Route Handling
 - Implement real-time route computation that adapts to changes in satellite positions and network conditions
 - Integrate variables such as satellite failures and varying link qualities to mimic real-world complexities

Thank You!